Day 162: Hypothesis Testing for ML, with a Real A/B Test

MD Mutasim Billah | 52-Week Data Science to ML/AI Roadmap | Week 24, Day 2 | Complete study guide

Watch: Hypothesis Testing and The Null Hypothesis, Clearly Explained!!! (StatQuest with Josh Starmer) | Power Analysis, Clearly Explained!!! (StatQuest with Josh Starmer)

Script: `day162_hypothesis_testing_ab.py` · every number in this guide is copied by program from `day162_run_output.txt`, the real output of that script.

Is a difference real, or could chance alone have made it? Today answers that on a real randomised experiment: 64,000 customers of a retailer, split at random into a men's email, a women's email and no email. The men's email raised visits by +72% and purchases by +119%, far beyond chance. Then the machinery itself is tested: with no effect at all it says 'significant' 0.0475 of the time, as promised, but checking the result 21 times and stopping at the first success raises that to 0.2430. Power calculations say how many customers an experiment needs, and simulation on the real data agrees. Finally, two ML models are compared on the same rows: McNemar's test finds a gap (p 1.11e-02) that a test ignoring the pairing misses (p 0.5409), but it also says 'different' too often when the only difference is the training sample.

0. Where today fits

DEFINITION — start here, no prior knowledge assumed

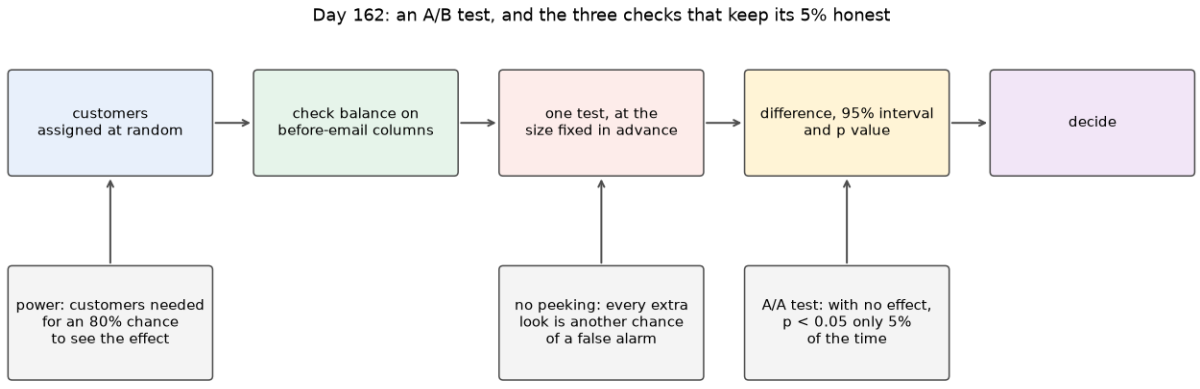
A **hypothesis test** asks whether data are surprising under a boring explanation, the **null hypothesis** (here: the email changes nothing). The **p value** is the chance of a result at least this extreme if the null hypothesis were true. Below a threshold fixed in advance (α, usually 0.05) the result is called **significant**.

Two mistakes are possible. A **false positive** (type I error): calling chance an effect; the test promises this happens at most α of the time when there is no effect. A **false negative** (type II error): missing a real effect. **Power** is the chance of not missing it, and it grows with the sample size.

REAL-WORLD SCENARIO

A shop sends a new email to half its customers and nothing to the other half. A week later the emailed half bought a little more. Did the email work, or did the emailed half happen to contain more keen shoppers? Because the halves were chosen at random, a hypothesis test can say how often chance alone would produce a gap this large.

SYSTEM DIAGRAM



An A/B test as a pipeline. The three grey boxes are the checks that keep its promised 5% false-positive rate true: power fixes the size in advance, no peeking, and an A/A test.

The data: Kevin Hillstrom's *MineThatData E-Mail Analytics Challenge* (2008). 64,000 customers who had bought in the past year were assigned at random to three groups of about 21,300: a men's merchandise email, a women's merchandise email, or no email. For two weeks the retailer recorded whether each customer **visited** the website, **bought** (conversion), and how much they **spent**. It is downloaded on first run to ~/week24_data, outside the repository. Section 6 returns to Day 161's UCI Adult models.

Today's part for the week's build

`day162_exercise.py` is today's toolkit (the data loader, the two-proportion and Welch tests, the permutation test, the sample-size and power formulas, and McNemar's test), each checked against a calculation done by hand. It is the second part of Day 167's report card.

ACTUAL OUTPUT

```
two_prop_test, 30/100 vs 20/100: by hand pooled 0.25, spread sqrt(0.25 x 0.75 x 2/100) = 0.0612, z
    = 0.1/0.0612 = 1.633, p = 0.1025; interval 0.1 +/- 1.96 x sqrt(0.21/100 + 0.16/100) =
    [-0.0192, 0.2192]
  got z 1.633, p 0.1025, interval [-0.0192, 0.2192]
welch_test, (1, 2, 3, 4) vs (2, 4, 6, 8): by hand variances 1.667 and 6.667, spread sqrt(1.667/4 +
    6.667/4) = 1.443, t = -2.5/1.443 = -1.732, normal p 0.0833 (with 4 rows a t table would give a
    larger p; the lesson's groups have thousands)
  got t -1.732, p 0.0833
permutation_test, (1, 2, 3) vs (4, 5, 6): by hand 20 ways to choose 3 of 6, and only 2 are as
    extreme ((1, 2, 3) and (4, 5, 6)), so p = 2/20 = 0.1
  got difference -3.0, p 0.1052 from 10000 shuffles (their own spread: sqrt(0.1 x 0.9 / 10000) =
      0.0030)
n_per_arm, 0.2 vs 0.1: by hand (1.960 x sqrt(2 x 0.15 x 0.85) + 0.842 x sqrt(0.16 + 0.09))^2 /
    0.1^2 = (0.9897 + 0.4208)^2 / 0.01 = 199.0
  got 199.0; power_two_prop at 199 per group: 0.8001 (by hand 0.80)
mcnemar, b = 10 rows only model 1 got wrong, c = 2 only model 2: by hand chi2 (|10 - 2| - 1)^2 /
    12 = 4.083, p 0.0433; exact 2 x (1 + 12 + 66) / 2^12 = 0.0386
  got b 10, c 2, chi2 4.083, p 0.0433, exact 0.0386
load_hillstrom: 64000 customers; Mens E-Mail 21307, Womens E-Mail 21387, No E-Mail 21306
```

How to read each section

Step	What it gives you
DEFINITION	What the idea is, in plain words. Start here.
REAL-WORLD SCENARIO	An everyday situation where the same idea shows up.
MATHEMATICS	The formula, when the section introduces one.
SPECIFICATION	The exact setup of the experiment.
IMPLEMENTATION	The real code from the script.
ACTUAL OUTPUT	What the script really printed, copied unchanged.
What it means	How to read the numbers, and what they do and do not show.
Why choose this? / Why NOT this?	When the idea is the right tool, and what it costs.

1. The experiment, and a check that the randomisation worked

DEFINITION — start here, no prior knowledge assumed

Random assignment is what makes the groups comparable: any difference after the emails can only come from the emails or from chance. That can be checked. The file records what was known *before* the emails (months since the last purchase, dollars spent last year, men's or women's buyer, new customer, channel, area); if the randomisation worked, these should differ between groups only by chance.

First draft: the loader split each line at commas, and failed: the history column contains values such as "$750 - $1,000", written in quotes because of the comma inside. Python's csv module reads the quotes correctly.

REAL-WORLD SCENARIO

Before trusting a drug trial, check that the treated and untreated patients had similar ages and health to begin with.

ACTUAL OUTPUT

```
Hillstrom's email experiment: 64000 customers, two weeks after the emails
  Mens E-Mail     21307 customers: visited 0.1828, bought 0.0125, spent $1.423 each
  Womens E-Mail   21387 customers: visited 0.1514, bought 0.0088, spent $1.077 each
  No E-Mail       21306 customers: visited 0.1062, bought 0.0057, spent $0.653 each
balance: 22 tests of before-email columns (11 columns x 2 emailed groups): 0 below 0.05 (chance alone expects 1.1); smallest
    p 0.159
largest standardised difference 0.0137 (a common rule of thumb worries above 0.1)
Actual output from this lesson's run.
```

What it means

Both emails raised all three outcomes. Visits: 0.1828 with the men's email, 0.1514 with the women's, 0.1062 with none. The balance check found nothing: none of the 22 tests fell below 0.05 (about 1 was expected by chance), and the largest standardised difference was 0.0137, far below the usual worry line of 0.1. Nothing here suggests the groups differed before the emails.

WHY CHOOSE THIS?

Cheap, and it catches broken randomisation (a bug in assignment, or a group filtered differently) before anyone reads the outcome.

WHY NOT THIS? (TRADEOFFS)

It can only check what was recorded; and with many columns, a few 'significant' imbalances are expected by chance alone.

2. Two-proportion z tests and Welch's test

DEFINITION — start here, no prior knowledge assumed

For a yes/no outcome (visited, bought), compare the two groups' rates. If there were no effect, both groups would share one rate, the **pooled** rate, and the difference between them would spread around 0 by a known amount. **z** counts how many of those spreads the observed difference is from 0, and the normal curve turns z into a p value.

For an amount (dollars spent), **Welch's test** does the same with the two groups' means, without assuming the groups have the same variance. Both report a **95% confidence interval**: the difference plus or minus 1.96 spreads.

REAL-WORLD SCENARIO

A newspaper poll says 52% against 48% with 'a margin of error of 3 points': that margin is the same spread.

MATHEMATICS

$$z = (p_1 - p_0) / \sqrt{(p_{pool}(1-p_{pool})(1/n_1 + 1/n_0))}, \quad p_{pool} = (x_1 + x_0)/(n_1 + n_0); \quad p\ value = P(|Z| \geq |z|)$$

$$95\%\ interval: (p_1 - p_0) \pm 1.96 \sqrt{(p_1(1-p_1)/n_1 + p_0(1-p_0)/n_0)}$$

$$Welch: t = (mean_1 - mean_0) / \sqrt{(s_1^2/n_1 + s_0^2/n_0)}$$

IMPLEMENTATION

```
def two_prop_test(x1, n1, x0, n0):
    p1, p0 = x1 / n1, x0 / n0
    pp = (x1 + x0) / (n1 + n0)
    se0 = math.sqrt(pp * (1 - pp) * (1 / n1 + 1 / n0))
    z = (p1 - p0) / se0 if se0 > 0 else 0.0
    se = math.sqrt(p1 * (1 - p1) / n1 + p0 * (1 - p0) / n0)
    return p1 - p0, z, p_two_sided(z), p1 - p0 - Z975 * se, p1 - p0 + Z975 * se
```

ACTUAL OUTPUT

```
  Mens E-Mail vs none, visit     : +0.0766 [+0.0700, +0.0832], z 22.49, p 5.7e-112 (+72% relative)
  Mens E-Mail vs none, conversion: +0.0068 [+0.0050, +0.0086], z 7.39, p 1.5e-13 (+119% relative)
  Mens E-Mail vs none, spend     : $+0.770 [+0.485, +1.055], t 5.30, p 1.2e-07 (+118% relative)
  Womens E-Mail vs none, visit     : +0.0452 [+0.0389, +0.0516], z 13.95, p 3.2e-44 (+43% relative)
  Womens E-Mail vs none, conversion: +0.0031 [+0.0015, +0.0047], z 3.78, p 1.6e-04 (+54% relative)
  Womens E-Mail vs none, spend     : $+0.424 [+0.169, +0.680], t 3.26, p 1.1e-03 (+65% relative)
men's email vs women's email, visit: +0.0314 [+0.0243, +0.0384], p 3.8e-18
men's email vs women's email, spend: $+0.345 [+0.033, +0.658], p 0.030
among buyers only (not randomised: the email changes who buys, so these groups are different people):
  Mens E-Mail     267 buyers, mean $113.53, median $66.98
  Womens E-Mail   189 buyers, mean $121.89, median $94.71
  No E-Mail       122 buyers, mean $114.00, median $88.37
Actual output from this lesson's run.
```

What it means

The men's email raised visits by +0.0766 (7.7 points, +72%), with z = 22.49: a p value of 5.7e-112, which chance essentially never produces. Purchases rose +119% and spend per customer by $0.770. The women's email worked too, less strongly (+43% visits). The two emails also differ from each other on spend, but only just: interval [+0.033, +0.658], p 0.030.

Among buyers only, spend per buyer was about the same in every group (mean $113.53, $121.89, $114.00): the emails seem to add buyers rather than bigger baskets. But that comparison is no longer randomised (the email changes *who* buys), so it describes the groups and proves nothing about the cause.

WHY CHOOSE THIS?

Fast, exact enough for thousands of customers, and the interval says how big the effect is, not only whether it exists.

WHY NOT THIS? (TRADEOFFS)

Relies on the normal approximation: fine with thousands of rows, doubtful with a few dozen or very rare events.

3. Permutation tests: the same question, no formula

DEFINITION — start here, no prior knowledge assumed

If the email did nothing, the group labels are arbitrary: any customer could have carried any label. So shuffle the labels at random, recompute the difference, and repeat 10,000 times. The p value is the share of shuffles that produce a difference at least as large as the real one. No normal curve is assumed.

REAL-WORLD SCENARIO

To test whether a coin is fair without any formula, flip a known fair coin the same number of times, many times over, and see how often it does as well.

IMPLEMENTATION

```
def permutation_test(a, b, R=R_PERM, seed=0):
    x = np.r_[a, b]; obs = a.mean() - b.mean()
    rng = np.random.RandomState(seed); hits = 0
    for _ in range(R):
        p = rng.permutation(len(x))
        hits += abs(x[p[:len(a)]].mean() - x[p[len(a):]].mean()) >= abs(obs) - 1e-12
    return obs, (hits + 1) / (R + 1)
```

ACTUAL OUTPUT

```
10000 shuffles each; formula's p against the shuffles' p
  bought, men's vs women's email: difference +0.0037; z test p 0.0002, permutation p 0.0001 (8 s)
    (the smallest p that 10000 shuffles can give is 1/10001 = 0.0001)
  spend, men's vs women's email: difference +0.3454; Welch p 0.0305, permutation p 0.0288 (7 s)
    (the shuffles' p carries its own sampling spread: about 0.0017 at this p)
  spend among buyers, women's email vs none: difference +7.8921; Welch p 0.5140, permutation p 0.5099 (0 s)
Actual output from this lesson's run.
```

What it means

The two methods agree. Spend, men's against women's email: Welch 0.0305, shuffles 0.0288, a gap the size of the shuffles' own sampling spread (0.0017). Spend among buyers: 0.5140 against 0.5099. For purchases, no shuffle out of 10,000 matched the real difference, so the shuffles can only say p is at most 1/10001 = 0.0001; the formula says 0.0002, which would need many more shuffles to confirm.

Spend is mostly zeros with a few large purchases, the kind of data where a normal approximation might fail. With about 21,000 customers per group it did not.

WHY CHOOSE THIS?

Assumption-light, works for any statistic (a median, a ratio, a model score), and is easy to explain.

WHY NOT THIS? (TRADEOFFS)

Slow (10,000 refits of the statistic), cannot resolve p values below 1/(R + 1), and its own p carries Monte Carlo noise.

4. A/A tests, and peeking

DEFINITION — start here, no prior knowledge assumed

An **A/A test** compares two groups that received the *same* treatment, so every 'significant' result is a false positive. Here the 21,306 untreated customers are split at random into two halves, 2,000 times. If the machinery is honest, $p < 0.05$ in about 5% of splits, and the p values are spread evenly between 0 and 1.

Peeking: checking the test as customers arrive (every 500 per group, 21 looks) and stopping the first time $p < 0.05$. Each look is one more chance for a false alarm.

REAL-WORLD SCENARIO

A product team watches its dashboard every morning and announces the new design a winner on the first day the number turns green.

ACTUAL OUTPUT

```
2000 A/A experiments: the 21306 untreated customers split at random into two groups of 10653
  visit     : p < 0.05 in 0.0475, p < 0.01 in 0.0065
  conversion: p < 0.05 in 0.0555, p < 0.01 in 0.0090
  spend     : p < 0.05 in 0.0490, p < 0.01 in 0.0080
peeking: a test after every 500 customers per group (21 looks), stopping at the first p < 0.05
  visit     : falsely 'significant' at some look 0.2430 (1 look 0.0470, 5 looks 0.1350, 10 looks 0.1860); at the last look
      alone 0.0475
  conversion: falsely 'significant' at some look 0.2445 (1 look 0.0480, 5 looks 0.1350, 10 looks 0.1870); at the last look
      alone 0.0555
Actual output from this lesson's run.
```

What it means

The machinery is honest: false positives in 0.0475 of A/A splits for visits, 0.0555 for purchases and 0.0490 for spend, all within chance of 5% (2,000 splits have a spread of about 0.005). Peeking breaks it: stopping at the first significant look gave 0.2430 false positives for visits and 0.2445 for purchases, about five times the promise. The damage grows with every look: 0.1350 after 5 looks, 0.1860 after 10.

The purchase p values bunch into some bins and leave others empty: with about 60 buyers per half, the counts are whole numbers, so only certain p values can occur. The 5% promise still held.

WHY CHOOSE THIS?

An A/A test checks the whole pipeline (assignment, logging, the test) on real data, where theory cannot.

WHY NOT THIS? (TRADEOFFS)

It costs traffic (or, as here, half a control group); fixing peeking needs sequential methods that adjust each look's threshold.

5. Power: how many customers does an experiment need?

DEFINITION — start here, no prior knowledge assumed

Before an experiment starts, choose the smallest effect worth detecting, then the sample size that gives a good chance (usually 80%) of detecting it. Too small and a real effect is likely to be missed; the experiment was wasted.

The formula is checked by simulation: draw n customers per group (with replacement) from the real groups, test, repeat 1,000 times, and count how often the test finds the effect.

REAL-WORLD SCENARIO

A clinical trial must enrol enough patients to have a fair chance of seeing the drug work, before the first patient is treated.

MATHEMATICS

$$n \text{ per group} = [z_{1-\alpha/2} \sqrt{(2\, p_{avg}(1-p_{avg})} + z_{power} \sqrt{(p_1(1-p_1) + p_0(1-p_0))}]^2 / (p_1 - p_0)^2$$

$$z_{0.975} = 1.96,\ z_{0.8} = 0.84;\ p_{avg} = (p_1 + p_0)/2$$

IMPLEMENTATION

```
def n_per_arm(p1, p0, alpha=ALPHA, power=0.8):
    za, zb = NormalDist().inv_cdf(1 - alpha / 2), NormalDist().inv_cdf(power)
    pb = (p1 + p0) / 2
    return (za * math.sqrt(2 * pb * (1 - pb)) + zb * math.sqrt(p1 * (1 - p1) + p0 * (1 - p0))) **
        2 / (p1 - p0) ** 2
```

ACTUAL OUTPUT

```
customers per group for 80% power at alpha 0.05, if the true rates are the ones observed:
  Mens E-Mail vs none, visit     : 0.1828 vs 0.1062 -> 330 per group
  Mens E-Mail vs none, conversion: 0.0125 vs 0.0057 -> 3066 per group
  Womens E-Mail vs none, visit     : 0.1514 vs 0.1062 -> 860 per group
  Womens E-Mail vs none, conversion: 0.0088 vs 0.0057 -> 11723 per group
checked by simulation: 1000 experiments of n per group, drawn with replacement from the real groups
  Mens E-Mail vs none, visit: n 100: 0.358 / 0.337; n 200: 0.632 / 0.587; n 300: 0.775 / 0.762; n 500: 0.929 / 0.932; n 800:
      0.996 / 0.992  (simulated / formula)
  Mens E-Mail vs none, conversion: n 500: 0.186 / 0.204; n 1000: 0.361 / 0.359; n 2000: 0.650 / 0.619; n 3000: 0.789 /
      0.792; n 4000: 0.899 / 0.893; n 6000: 0.980 / 0.975; n 8000: 0.998 / 0.995  (simulated / formula)
  Womens E-Mail vs none, conversion: n 2000: 0.195 / 0.211; n 5000: 0.441 / 0.448; n 10000: 0.743 / 0.735; n 15000: 0.879 /
      0.887; n 20000: 0.961 / 0.955  (simulated / formula)
Actual output from this lesson's run.
```

What it means

Small effects on rare outcomes need large experiments. The men's email's effect on visits needs only 330 per group; its effect on purchases 3066 per group; the women's email's smaller effect on purchases 11723 per group. Hillstrom's 21,000 per group was enough for all four.

Simulation agrees with the formula: at 3,000 customers per group the men's purchase effect was found in 0.789 of simulated experiments against the formula's 0.792; at 10,000 the women's in 0.743 against 0.735.

WHY CHOOSE THIS?

Says in advance whether an experiment can answer its question, and stops underpowered tests that will mislead.

WHY NOT THIS? (TRADEOFFS)

Needs a guess of the effect size before seeing data; here the observed rates were used, which only an analysis afterwards can do.

6. Comparing two models on the same rows: McNemar's test

DEFINITION — start here, no prior knowledge assumed

Two classifiers are scored on the same 6,032 validation rows. Rows both get right, or both get wrong, say nothing about which is better. Only the disagreements matter: **b** rows only the first model gets wrong, **c** rows only the second. If the models were equally good, each disagreement would be a coin flip, so b and c should be close. McNemar's test asks whether they are too far apart for coin flips.

As a check of what it means, the same procedure is trained twice on two random halves of the fitting rows, 100 times: the procedures are identical, only the training rows differ.

REAL-WORLD SCENARIO

Two doctors read the same 6,000 X-rays. What matters is the films where they disagree, and which of them was right on those.

MATHEMATICS

$$\chi^2 = (|b - c| - 1)^2 / (b + c), \quad 1 \text{ degree of freedom}; \quad \text{exact: } 2 \cdot P(X \le \min(b, c)),\ X \sim \text{Binomial}(b + c, \tfrac{1}{2})$$

IMPLEMENTATION

```
def mcnemar(e1, e2):
    b = int(np.sum(e1 & ~e2)); c = int(np.sum(~e1 & e2))
    chi2 = (abs(b - c) - 1) ** 2 / (b + c) if b + c else 0.0
    n, k = b + c, min(b, c)
    lp = [math.lgamma(n + 1) - math.lgamma(i + 1) - math.lgamma(n - i + 1) - n * math.log(2) for i
        in range(k + 1)]
    m = max(lp) if lp else 0.0
    exact = min(1.0, 2 * math.exp(m) * sum(math.exp(v - m) for v in lp)) if n else 1.0
    return b, c, chi2, math.erfc(math.sqrt(chi2 / 2)), exact
```

ACTUAL OUTPUT

```
validation rows 6032; error rates: logistic 0.1592, boost 100 0.1376, boost 200 0.1338
  logistic vs boost 200: only logistic wrong 279, only boost 200 wrong 126; McNemar chi2 57.05, p 4.26e-14 (exact 2.09e-14);
      unpaired z test p 0.0001
  boost 100 vs boost 200: only boost 100 wrong 49, only boost 200 wrong 26; McNemar chi2 6.45, p 1.11e-02 (exact 1.06e-02);
      unpaired z test p 0.5409
same procedure, different training rows: 100 times, the fitting rows cut into two random halves,
one model trained on each, McNemar on the validation rows (no real difference in the procedure):
  logistic : p < 0.05 in 0.09 of 100 (a test of fixed models expects 0.05; spread of that share about 0.022) (17 s)
  deep tree: p < 0.05 in 0.14 of 100 (a test of fixed models expects 0.05; spread of that share about 0.022) (131 s)
Actual output from this lesson's run.
```

What it means

Boosting beats logistic regression clearly: 279 rows only logistic regression got wrong against 126 only boosting got wrong, p 4.26e-14. The useful case is the close one. 200 boosting rounds against 100: error 0.1338 against 0.1376, a gap of 23 rows. McNemar gives p 1.11e-02; a two-proportion test that treats the two error rates as coming from different people gives 0.5409. Pairing removes the rows both models handle the same way, which is most of the noise.

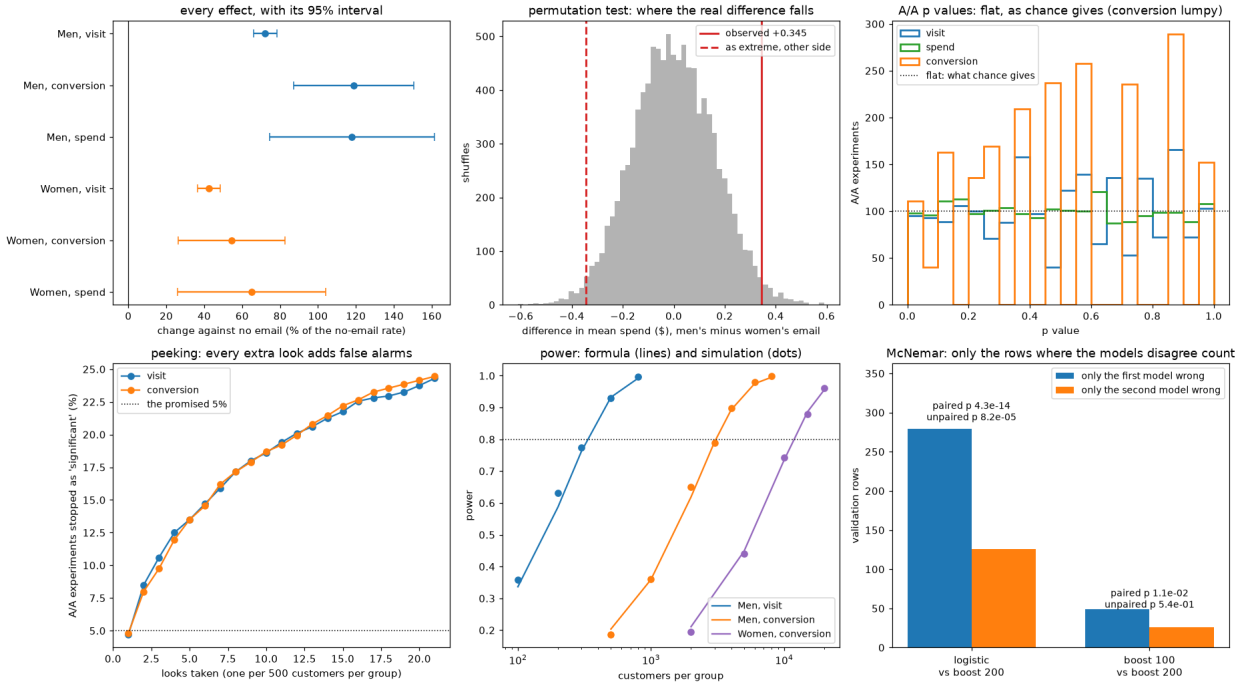
But the control shows what McNemar does not measure. Two models from the *same* procedure on different training rows were called different in 0.09 of 100 repeats for logistic regression and 0.14 for the deep tree, against the 0.05 a test of fixed models promises (the spread of such a share is about 0.022). McNemar compares two fitted models; it ignores that a new training sample would have given different models. The unstable tree suffers most. A test of training procedures needs the training itself repeated on resampled data.

WHY CHOOSE THIS?

Uses the pairing, so it detects small differences between two fixed models that an unpaired test misses.

WHY NOT THIS? (TRADEOFFS)

Treats the two models as fixed: it says nothing about whether the training procedure is better, and over-rejects for unstable models.



Top: every email effect with its 95% interval, as a share of the no-email rate; the permutation null for spend (men's against women's email) with the real difference in red; A/A p values, flat for visits and spend, lumpy for purchases. Bottom: false positives grow with every look; power by sample size, formula (lines) against simulation (dots); McNemar's disagreement counts, with paired and unpaired p values.

Interview preparation — technical questions

Q: How would you analyse a conversion-rate A/B test?

A: A two-proportion z test with a 95% interval on the difference, after checking balance on pre-experiment columns; a permutation test if the counts are small.

Q: What is an A/A test for?

A: To check the pipeline: with no treatment, $p < 0.05$ should happen about 5% of the time. Here it was 4.75% to 5.55%.

Q: Why is peeking a problem?

A: Each look is another chance of a false positive. Checking 21 times and stopping at the first $p < 0.05$ gave 24% false positives instead of 5%.

Q: How many users do you need?

A: Fix α, power and the smallest effect worth finding, then use the sample-size formula. A rare outcome with a small lift (0.57% to 0.88%) needed 11,723 per group.

Q: How do you test whether one classifier beats another on a test set?

A: McNemar's test on the rows where they disagree. It found 200 against 100 boosting rounds different (p 0.011) where an unpaired test did not (p 0.54).

Interview preparation — theoretical questions

Q: What does a p value not tell you?

A: Not the chance the null hypothesis is true, and not the size of the effect: that is the interval's job.

Q: Why does randomisation matter?

A: It makes the groups comparable in everything, recorded or not, so a difference can only come from the treatment or chance.

Q: Permutation test or z test?

A: They agreed here with 21,000 per group. Permutation needs fewer assumptions; the formula is faster and resolves tiny p values.

Q: What does McNemar's test not cover?

A: Variation from the training sample. Identical procedures on different halves of the data were called different 9% (logistic) and 14% (deep tree) of the time.

Key takeaway and what's next

> **KEY TAKEAWAY**
>
> A p value is only as honest as the procedure around it. On a real randomised experiment the tests worked as promised: A/A splits gave about 5% false positives and permutation tests agreed with the formulas. Peeking broke the promise fivefold, power showed that rare outcomes need large experiments, and McNemar's test showed both the strength of pairing and the limit of testing two fixed models.

- Men's email: visits +72%, purchases +119%; randomisation check: none of 22 tests below 0.05.
- A/A false positives 0.0475 to 0.0555; with 21 looks, 0.2430.
- Sample sizes for 80% power: 330 per group to 11723 per group; simulation within a few points of the formula.
- McNemar: p 1.11e-02 where an unpaired test gave 0.5409; same-procedure control 0.09 and 0.14 against 0.05.

Suggested watch order, if starting from zero

- 1. Hypothesis Testing and The Null Hypothesis, Clearly Explained!!! (StatQuest with Josh Starmer) — null hypotheses and p values, with simple pictures.
- 2. Power Analysis, Clearly Explained!!! (StatQuest with Josh Starmer) — how to choose a sample size before an experiment, and why that prevents p-hacking.

Day 163 preview

- **Confidence intervals and the bootstrap.** Today's intervals came from formulas that assume large, well-behaved samples. Tomorrow resamples the data itself, and checks how often each kind of interval really contains the truth.